

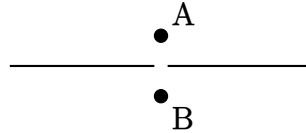
§5 Exercises

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5.1. The real line with two origins

Let A and B be two points not on the real line $\mathbb{R}$. Consider the set $S = (\mathbb{R} - \{0\}) \cup \{A, B\}$.



For any two positive real numbers c, d , define

$$I_A(-c, d) = (-c, 0) \cup \{A\} \cup (0, d)$$

and similarly for $I_B(-c, d)$, with B instead of A . Define a topology on S as follows: On $(\mathbb{R} - \{0\})$, use the subspace topology inherited from $\mathbb{R}$, with open intervals as a basis. A basis of neighborhoods at A is the set $\{I_A(-c, d) \mid c, d > 0\}$; similarly, a basis of neighborhoods of B is $\{I_B(-c, d) \mid c, d > 0\}$.

- (a) Prove that the map $h : I_A(-c, d) \rightarrow (-c, d)$ defined by

$$\begin{aligned} h(x) &= x \text{ for } x \in (-c, 0) \cup (0, d), \\ h(A) &= 0 \end{aligned}$$

is a homeomorphism.

- (b) Show that S is locally Euclidean and second countable, but not Hausdorff.

Proof. (a) Given $x \in (-c, d)$ where $x \neq 0$, choose a neighborhood N to be $(-c, 0)$ if $x < 0$ and $(0, d)$ otherwise, which pulls back to $(-c, 0)$ and $(0, d)$ respectively. If $x = 0$, then the pullback of a neighborhood $(-c, d)$ is $I_A(-c, d)$. In either case, the pullback is open, so h is continuous.

Concluding with h^{-1} , the case where $x \neq A$ is nearly identical to the above, and if $x = A$, then take a neighborhood $I_A(-c, d)$, which pulls back to $(-c, d)$, which is open.

- (b) The local Euclidean condition is proven by (a), along with an entirely analagous statement for B . The basis $\{I_A(-c, d)\} \cup \{I_B(-c, d)\}$ for rational c and d is countable. Finally, if S were Hausdorff, there would be a neighborhood U of A and a neighborhood V of B such that $U \cap V = \emptyset$. Then we would have $a, b, c, d \in \mathbb{R}^+$ such

that $I_A(-a, b) \subset U$ and $I_B(-c, d) \subset V$ are disjoint, but this is clearly false since $I_A(-a, b) \cap I_B(-c, d) \supset (\min(-a, -c), \max(-a, -c))$. □

5.2. A sphere with a hair

A fundamental theorem of topology, the theorem on invariance of dimension, states that if two nonempty open sets $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ are homeomorphic, then $n = m$. Use the idea of Example 5.4 [“the cross”] as well as the theorem on invariance of dimension to prove that the sphere with a hair in $\mathbb{R}^3$ is not locally Euclidean at q . Hence it cannot be a topological manifold.

Proof. Assume that there is a neighborhood N of q which admits a homeomorphism ϕ onto some open ball $B \subset \mathbb{R}^n$ such that $\phi(q) = 0$. Then $N - \{q\}$ has two connected components, so n cannot be greater than 1, but it also cannot be 0, so $n = 1$. Each component of $B - \{0\}$ is a 1-manifold, however, while $N - \{q\}$ has a 1-manifold as one component and a 2-manifold as the other component, so invariance of dimension implies that ϕ is not a homeomorphism. □

5.3. Charts on a sphere

Let S^2 be the unit sphere

$$x^2 + y^2 + z^2 = 1$$

in $\mathbb{R}^3$. Define in S^2 the six charts corresponding to the six hemispheres—the front, rear, right, left, upper, and lower hemispheres:

$$\begin{aligned} U_1 &= \{(x, y, z) \in S^2 \mid x > 0\}, & \phi_1(x, y, z) &= (y, z), \\ U_2 &= \{(x, y, z) \in S^2 \mid x < 0\}, & \phi_2(x, y, z) &= (y, z), \\ U_3 &= \{(x, y, z) \in S^2 \mid y > 0\}, & \phi_3(x, y, z) &= (x, z), \\ U_4 &= \{(x, y, z) \in S^2 \mid y < 0\}, & \phi_4(x, y, z) &= (x, z), \\ U_5 &= \{(x, y, z) \in S^2 \mid z > 0\}, & \phi_5(x, y, z) &= (x, y), \\ U_6 &= \{(x, y, z) \in S^2 \mid z < 0\}, & \phi_6(x, y, z) &= (x, y). \end{aligned}$$

Describe the domain $\phi_4(U_{14})$ of $\phi_1 \circ \phi_4^{-1}$ and show that $\phi_1 \circ \phi_4^{-1}$ is C^∞ on $\phi_4(U_{14})$. Do the same for $\phi_6 \circ \phi_1^{-1}$.

Proof. Since U_{14} is a quarter-sphere, namely the one where $x > 0$ and $y < 0$, $\phi_4(U_{14})$ is the projection of U_{14} onto the x, z -plane, which is the open half-ball $B_1 = \{(x, z) : x > 0, x^2 + z^2 < 1\} \in \mathbb{R}^2$. Explicitly, if $(x, z) \in B_1$,

$$\begin{aligned} \phi_1(\phi_4^{-1}(x, z)) &= \phi_1(x, \sqrt{1 - x^2 - z^2}, z) \\ &= (\sqrt{1 - x^2 - z^2}, z), \end{aligned}$$

which is C^∞ on B_1 .

Similarly, skipping some details much like those above, we see that $B_2\phi_1(U_{61}) = \{(y, z) : z < 0, y^2 + z^2 < 1\}$, and for $x \in B_2$,

$$\phi_6(\phi_1^{-1}(y, z)) = \left(\sqrt{1 - y^2 - z^2}, y \right),$$

which is C^∞ on B_2 . □

5.4. Existence of a coordinate neighborhood

Let $\{(U_\alpha, \phi_\alpha)\}$ be the maximal atlas on a manifold M . For any open set U in M and a point $p \in U$, prove the existence of a coordinate open set U_α such that $p \in U_\alpha \subset U$.

Proof. Let $\mathcal{A} = \{(U_\alpha, \phi_\alpha)\}$. Since $\mathcal{A}$ is an atlas, there exists $(U_1, \phi_1) \in \mathcal{A}$ such that $p \in U_1$. Set $U_2 = U_1 \cap U$. Then U_2 is open, $\phi_2 = \phi_1|_{U_2}$ is a homeomorphism, and (U_1, ϕ_1) is smoothly compatible with (U_2, ϕ_2) because $U_2 \cap U_1 = U_2$ and $\phi_1 = \phi_2$ on U_2 , so $\phi_1 \circ \phi_2^{-1} = \phi_2 \circ \phi_1^{-1} = I$ where I is the identity function on $\phi_1(U_2)$.

In fact, given any $(U_\alpha, \phi_\alpha) \in \mathcal{A}$, $\phi_\alpha \circ \phi_2^{-1}$ is defined on $\phi_2(U_\alpha \cap U_2)$, and since $\phi_2 = \phi_1$ on any subset of U_2 , $\phi_\alpha \circ \phi_2^{-1} = \phi_\alpha \circ \phi_1^{-1}$ and likewise $\phi_2 \circ \phi_\alpha^{-1} = \phi_1 \circ \phi_\alpha^{-1}$, so the smooth compatibility of (U_α, ϕ_α) with (U_2, ϕ_2) is a result of its compatibility with (U_1, ϕ_1) .

Thus, (U_2, ϕ_2) is compatible with every chart in $\mathcal{A}$, so since $\mathcal{A}$ is maximal, $(U_2, \phi_2) \in \mathcal{A}$. $\square$

5.5. An atlas for a product manifold

Prove Proposition 5.18: If $\{(U_\alpha, \phi_\alpha)\}$ and $\{(V_i, \psi_i)\}$ are C^∞ atlases for the manifolds M and N of dimensions m and n , respectively, then the collection

$$\{(U_\alpha \times V_i, \phi_\alpha \times \psi_i) : U_\alpha \times V_i \rightarrow \mathbb{R}^m \times \mathbb{R}^n\}$$

of charts is a C^∞ atlas on $M \times N$. Therefore, $M \times N$ is a C^∞ manifold of dimension $m + n$.

Proof. $\square$